

✓ Setup system

Setup the system and get files from the internet.

In the following code, we will download the necessary files from the internet.

```
1 # SETUP: Please ignore this part.
2
3 # Remove existing files
4 import os
5
6 # Stop pygame from opening a window
7 os.environ["SDL_VIDEODRIVER"] = "dummy"
8
9 # Install kaleido package
10 !pip install -U kaleido==0.2.1
11
12 # Mount the drive:
13 from google.colab import drive
14 drive.mount('/content/drive')
15
16 try:
17     os.chdir("/content/drive/MyDrive/Race-2025")
18     print("Directory changed successfully.")
19 except FileNotFoundError:
20     print("Error: Race-2025 directory not found.")
21
22 # Create a local directory
23 !mkdir /content/drive/MyDrive/Race-2025
24
25 # Get the repo as a zip file and save it locally:
26 !wget https://github.com/pattichis/GraphSpeeds2/archive/refs/head
27 !unzip repo.zip -d /content/drive/MyDrive/Race-2025/
28 !rm -f repo.zip
29
30
31 #!find './' -maxdepth 1 -type f -delete
32
33 # Use wget with -nc to avoid overwriting local copy.
34 !rm -f lineart_v4.py*
35 !wget https://raw.githubusercontent.com/pattichis/GraphSpeeds2/main
36
37 # Download the pictures:
38 !wget -nc https://raw.githubusercontent.com/pattichis/GraphSpeeds2/
39 !wget -nc https://raw.githubusercontent.com/pattichis/GraphSpeeds2/
40 !wget -nc https://raw.githubusercontent.com/pattichis/GraphSpeeds2/
41 !wget -nc https://raw.githubusercontent.com/pattichis/GraphSpeeds2/
```

```

42
43 # Import the functions to use
44 from lineart_v4 import table, simulationVideo, textImage, plotTable

```

Requirement already satisfied: kaleido==0.2.1 in /usr/local/lib/python3.1
Drive already mounted at /content/drive; to attempt to forcibly remount,
Directory changed successfully.

--2026-08-20 18:24:25-- https://raw.githubusercontent.com/pattichis/GraphSpeeds2-main/main/lineart_v4.py
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.19
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.1
HTTP request sent, awaiting response... 200 OK
Length: 34249 (33K) [text/plain]
Saving to: 'lineart_v4.py'

lineart_v4.py 100%[=====] 33.45K --.-KB/s in 0.

2026-08-20 18:24:25 (1.84 MB/s) - 'lineart_v4.py' saved [34249/34249]

File 'Elephant.png' already there; not retrieving.

--2026-08-20 18:24:25-- <https://raw.githubusercontent.com/pattichis/GraphSpeeds2-main/main/Tortoise.jpg>
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.19
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.1
HTTP request sent, awaiting response... 404 Not Found
2026-08-20 18:24:26 ERROR 404: Not Found.

File 'Tortoise.jpg' already there; not retrieving.

File 'Koala.jpeg' already there; not retrieving.

Make videos I: Pick Find character images

Assignment: Find character images

1. Click on your Folder icon on the Left Hand Side.
2. Click on drive.
3. Click on MyDrive.
4. Click on Race-2025.
5. Click on GraphSpeeds2-main.
6. Look at the different images.
7. Write down the names of the image files that you want to use.

✓ Make videos II: Create tables for your characters

✓ Assignment: Create the tables and equations for your characters

1. Enter time and distance units in column_labels.
2. Enter time_values and distance_values for each value.
3. Enter names and equations for each character.

```

1 # Prepare the labels for everyone
2 column_labels = ["Time: x (seconds)", "Distance: y (feet)"] # <- FI
3
4 # Table #1 values:
5 # normal waalking st
6 time_values      = [0, 1, 2, 5.50, 11] # <- ENTER x values
7 distance_values = [0, 4.54, 9.091, 25, 50] # <- ENTER y values
8
9 # Create your table:
10 char1 = table(x=time_values, y=distance_values, column_labels=column_labels)
11 char1.plotTable()
12 char1.saveImage("char1.png")
13
14
15 # Table #2 values:
16 # walking fast Beac
17 time_values      = [0, 1, 2, 4.5, 9 ] # <- ENTER x values
18 distance_values = [0, 5.56,11.1,25,50 ] # <- ENTER y values
19
20 # Create your table:
21 char2 = table(x=time_values, y=distance_values, column_labels=column_labels)
22 char2.plotTable()
23 char2.saveImage("char2.png")
24
25
26
27 # Table #3 values:
28 #jogging no se
29 time_values      = [0, 1, 2, 3.01, 6] # <- ENTER x values
30 distance_values = [0, 8.3, 16.6, 25, 50] # <- ENTER y values
31
32 # Create your table:
33 char3 = table(x=time_values, y=distance_values, column_labels=column_labels)
34 char3.plotTable()
35 char3.saveImage("char3.png")
36
37
38 # Table #4 values:
39 # sprinting Mcguire
40 time_values      = [0, 1, 2, 2.5, 5] # <- ENTER x values
41 distance_values = [0, 10, 20, 25, 50] # <- ENTER y values

```

```
42
43 # Create your table:
44 char4 = table(x=time_values, y=distance_values, column_labels=column_labels)
45 char4.plotTable()
46
47 char4.saveImage("char4.png")
48
49
50 # Merge all of the tables
51 tables_list = [char1, char2, char3, char4]
52
53 # Line equations of tables
54 legend_labels = [
55     'st: y = 4.54*x', # <- Modify name and the equation here.
56     'Beac: y = 5.56*x', # <- Modify name and the equation here.
57     'no se: y = 8.3*x', # <- Modify name and the equation here.
58     'Mcguire: y = 10*x'] # <- Modify name and the equation here.
59
60 # Plot tables
61 plotTablesLines(tables_list,
62                 fig_title = 'The Race', # title on top
63                 x_label = 'Elapsed time (seconds)', # appears along x-axis
64                 y_label = 'Distance in feet', # appears along y-axis
65                 legend_title = 'Journey', # title on the legend
66                 legend_labels = legend_labels, # text descriptions
67                 img_name = 'race.png') # Saves under img_name
```


Time: x (seconds)	Distance: y (feet)
0.00000	0.00000
1.00000	4.54000
2.00000	9.09100
5.50000	25.00000
11.00000	50.00000

Wrote char1.png

Time: x (seconds)	Distance: y (feet)
0.00000	0.00000
1.00000	5.56000
2.00000	11.10000
4.50000	25.00000
9.00000	50.00000

Wrote char2.png

Time: x (seconds)	Distance: y (feet)
0.00000	0.00000
1.00000	8.30000
2.00000	16.60000
3.01000	25.00000
6.00000	50.00000

Wrote char3.png

Time: x (seconds)	Distance: y (feet)
0.00000	0.00000
1.00000	10.00000
2.00000	20.00000
2.50000	25.00000
5.00000	50.00000

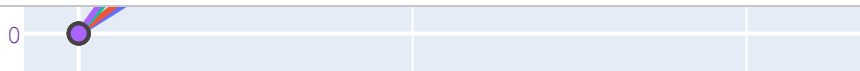
✓ Make videos III: Specify names, images, and speeds

Assignment: place your characters

1. ENTER the names of the characters for your video.
2. ENTER filenames for each character in your video.
3. ENTER the speeds for your characters.
Wrote char4.png

```

1 # Enter the names of your characters:
2 names = ["Skibidi toilet", "bacon egg and cheese", "no se", "Mcguir
3
4 # Enter the file names of your images:
5 images = ["snoopy.jpg", "red-race-car.jpg", "hello-kitty-2.png", "Cu
6
7 # Enter the speeds:
8 speeds = [4.59, 5.56, 8.3, 10]
9
10
11 # Directory:
12 directory = "/content/drive/MyDrive/Race-2025/GraphSpeeds2-main/"
13 images_list = [directory + filename for filename in images]
14
15 # Save all of the data:
16 data = [names, images_list, speeds]
```



✓ Make videos IV: Make and experiment with your video

Elapsed time (seconds)

Your video will be stored to the video file race.mp4 or a filename that you choose.

To make a video, one time unit (e.g., one hour) corresponds to one second of video.

If you set the `simulation_speed=2.0`, then one second will correspond to two time units (e.g., two hours).

For each second, the video contains a number of images. This is controlled by `fps` which means number of frames per second.

Assignment: experiment with your video

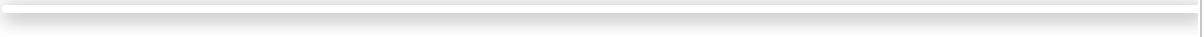
1. Enter the race distance, duration of the race, and video title.
2. Enter the units.
3. Change `simulation_speed` to 2.0. What do you observe?
4. Change `fps` to 20. What do you observe?

5. Save the video to a different file name.

```
1 # Experiment with the video
2 sim_video = simulationVideo(data = data,           # characters
3                             duration = 11,         # ENTER duration
4                             race_distance = 50,     # ENTER race dis
5                             vid_title = "Race")     # video title
6
7 sim_video.set_video(video_name="race.mp4",         # new filename
8                    fps=10,                          # frames per sec
9                    simulation_speed=1.0)            # speed of the s
10
11 sim_video.set_units(distance_string="feet",         # ENTER distance
12                    time_string="seconds",           # ENTER time uni
13                    speed_string="feet/second")      # ENTER speed un
14
15
16 # Create and save your video:
17 video = sim_video.create_video()
18 video.ipython_display()
```

```
File 'race.mp4' has been removed.  
VideoWriter initialized successfully.  
video file = race.mp4 closed.  
Moviepy - Building video __temp__.mp4.  
Moviepy - Writing video __temp__.mp4  
  
Moviepy - Done !  
Moviepy - video ready __temp__.mp4
```

0:00 / 0:11



✓ Make videos V: Make title and explanatory text images

You can create many text frames and save them into different frames.

This is done using `saveImage("filename.png")` as given below:

```
text = ""  
First line.  
Second line.
```

```
Third line. """"  
textImage(frame1.png", text)
```

Assignment: Create text images

Fill out the explanatory text images.

```
1 # Create Title and group name here  
2 text = """"Session 2 Group B:  
3 Team bacon egg and cheese  
4 """"  
5 textImage("frame1.png", text)  
6  
7  
8 # Who is racing?  
9 text = """"The  
10 - bacon egg and cheese,  
11 - skibidi toilet,  
12 - no se,  
13 - Mcguire  
14 are racing.  
15 """"  
16 textImage("frame2.png", text)  
17  
18  
19 # Why they are racing?  
20 text = """"They are racing because they wanna see  
21 who can cover 50 feet the fastest""""  
22 textImage("frame3.png", text)  
23  
24  
25 # Character 1 equation and explanation:  
26 text = """"Character 1 Table:  
27 For equation  $y = 4.54 * x$   
28 """"  
29 textImage("frame4.png", text)  
30  
31  
32 # Character 1 equation and explanation:  
33 text = """"Character 1 equation:  
34  $y = 4.54 * x$   
35 walking normal made me have a  
36 constant speed of 4.59ft. per second.  
37 I got 4.54 ft because  $50/11 = 4.54\text{ft}$   
38 """"  
39 textImage("frame5.png", text)  
40  
41  
42 # Character 2 equation and explanation:
```

```
43 text = """"Character 2 Table:
44 For equation  $y = 5.56 * x$ 
45 """"
46 textImage("frame6.png", text)
47
48
49 # Character 2 equation and explanation:
50 text = """"Character 2 equation:
51  $y = 5.56 * x$ 
52 walking fast made me have a
53 constant speed of 5.56ft per second.
54 I got 5.56ft because  $50/9 = 5.56\text{ft}$ 
55 """"
56 textImage("frame7.png", text)
57
58
59 # Character 3 equation and explanation:
60 text = """"Character 3 Table:
61 For equation  $y = 8.3 * x$ 
62 """"
63 textImage("frame8.png", text)
64
65
66 # Character 3 equation and explanation:
67 text = """"Character 3 equation:
68  $y = 8.3 * x$ 
69 jogging made me have a
70 constant speed of 8.3ft per second.
71 I got 8.3ft because  $50/6 = 8.3\text{ft}$ 
72 """"
73 textImage("frame9.png", text)
74
75
76 # Character 4 equation and explanation:
77 text = """"Character 4 Table:
78 For equation  $y = 10 * x$ 
79 """"
80 textImage("frame10.png", text)
81
82
83 # Character 4 equation and explanation:
84 text = """"Character 4 equation:
85  $y = 10 * x$ 
86 sprinting made me have a
87 constant speed of 10ft per second
88 I got 10ft because  $50/5 = 10\text{ft}$ 
89 """"
90 textImage("frame11.png", text)
91
92
93 # Explain the winner:
```

```

94 text = """"Next is comparison of race results
95 """"
96 textImage("frame12.png", text)
97
98
99 # Explain the winner:
100 text = """"Mcguire was the fastest
101 and won because he was
102 sprinting 10ft per second!
103 """"
104 textImage("frame13.png", text)
105
106
107 # Victory page:
108 text = """"we all won mcdonalds for racing!""""

```

✓ Make videos VII: Make the final video

You can make the final video by combining images and videos together.

Assignment: Make final video

Verify that your video satisfies the requirements of your assignment!

```

1 # Confirm that you have all the image here:
2 img_list = ["frame1.png", "frame2.png", "frame3.png",
3             "race.mp4",
4             "frame4.png", "char1.png", "frame5.png",
5             "char2.png", "frame6.png",
6             "char3.png", "frame7.png",
7             "char4.png", "frame8.png", "race.png",
8             "frame9.png", "frame10.png"]
9
10 # Create the video:
11 final_video=CreateVideo(video_name="all.mp4",      # This is file nam
12                          file_list=img_list,      # This the list of image
13                          fps=10,                  # The number of frames p
14                          duration=3.5)            # The duration for eac
15
16 final_video.ipython_display(maxduration=300)
17
18 # Display the video:
19 video.ipython_display()

```

